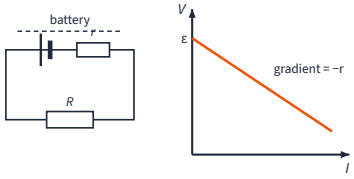
CHAPTER 16

# DIRECT CURRENT CIRCUITS

**Key idea:** charge is conserved at junctions and energy around loops.

## 1 E.M.F. & INTERNAL RESISTANCE

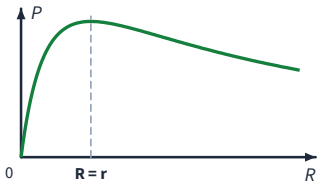



$$E = V + Ir$$

$$E = I(R + r)$$

**E.m.f.:** energy converted to electrical energy per unit charge. $V$–$I$ graph: intercept $E$, gradient $-r$.

## 2 POWER TRANSFER

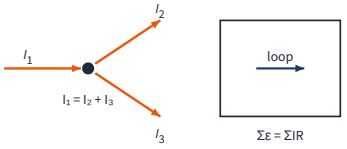



$$P = \frac{E^2 R}{(R+r)^2}$$

$$P_{\max} = \frac{E^2}{4r}$$

Maximum power in $R$ when $R = r$ (efficiency 50%).

## 3 KIRCHHOFF'S LAWS




$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

$$\sum E = \sum IR$$

- 1st law: conservation of **charge**
- 2nd law: conservation of **energy**
- Negative answer: current flows the other way

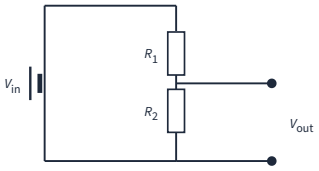
## 4 RESISTOR NETWORKS

$$R_s = R_1 + R_2$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

- Series: same $I$; parallel: same $V$
- Voltmeter across $R$: $\dfrac{RR_V}{R + R_V}$ (reads low)
- Cells in series: add $E$ and $r$

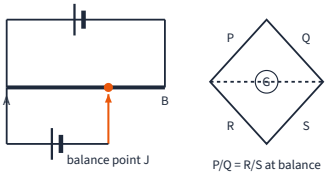
## 5 POTENTIAL DIVIDER




$$V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}$$

- Load across $R_2$: output falls
- Thermistor (NTC): $R \downarrow$ as $T \uparrow$; LDR: $R \downarrow$ as light $\uparrow$

## 6 POTENTIOMETER & BRIDGE




$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$\frac{P}{Q} = \frac{R}{S}$$

At balance no current in the galvanometer, so the potentiometer reads the true e.m.f.

## 7 METHOD FOR NETWORKS

1. Label every current with a direction.
2. Junction rule at each node.
3. Loop rule round enough loops, with signs.
4. Solve the simultaneous equations; check one loop.

## 8 COMMON SECTION A TRAPS

- Taking the terminal p.d. as the e.m.f. when current flows.
- Forgetting $r$ in $I = E/(R + r)$.
- Sign of an e.m.f. met from + to − in a loop.
- Balance length is measured from the **same** end as the driver's positive.

## EXAM TIPS

- ✓Redraw the circuit simply before calculating.
- ✓State "at balance no current flows through G".
- ✓Use ratios of lengths for potentiometers.
- ✓Check power: $EI = I^2R + I^2r$.